Chapter 5. Neural Networks and Backpropagation

Basic concepts of neural networks, multilayer perceptrons, backpropagation

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1 From Hand-Designed Features to Learned Functions

A classifier maps an observation $\mathbf{x}$ to an output $y = f(\mathbf{x}; \boldsymbol{\theta})$. Classical pipelines fix features by hand (color histograms, filter banks) and learn only a final classifier; deep learning trains the representation and the classifier end to end from labeled data. Two properties of features recur: a feature is *invariant* to a transformation if it does not change, and *equivariant* if it changes in the same way (Chapter 7).

2 Neurons, Layers and Shapes

Definition 5.1: multilayer perceptron

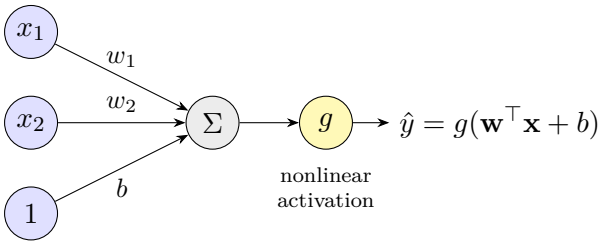
A neuron computes $h = \sigma(\mathbf{w}^\top \mathbf{x} + b)$. A fully connected layer stacks neurons:

$$\mathbf{z}^{(\ell)} = W_\ell \mathbf{a}^{(\ell-1)} + \mathbf{b}_\ell, \qquad \mathbf{a}^{(\ell)} = \sigma(\mathbf{z}^{(\ell)}), \qquad \mathbf{a}^{(0)} = \mathbf{x},$$

with $W_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$, $\mathbf{b}_\ell \in \mathbb{R}^{n_\ell}$, and σ applied elementwise. An MLP is a composition of such layers; a 1000-neuron layer is one layer, not 1000 layers.

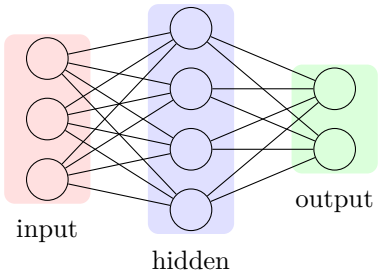
Example 5.2: parameter count

A two-layer classifier $\mathbf{s} = W_2 \,\mathrm{ReLU}(W_1 \mathbf{x})$ on CIFAR-10 images ($32 \times 32 \times 3 = 3072$ inputs), hidden width 100, 10 classes: $W_1 \in \mathbb{R}^{100 \times 3072}$, $W_2 \in \mathbb{R}^{10 \times 100}$, so $307{,}200 + 1000 = 308{,}200$ weights (plus 110 biases).



A single neuron: an affine combination of the inputs followed by a nonlinear activation g.

$$f(\mathbf{x}) = \mathrm{softmax}\left(W_2 \max(0, W_1 \mathbf{x})\right)$$



A two-layer network: every unit of a layer is connected to every unit of the next.

3 Why Nonlinearity and Depth

Without activations layers collapse: $W_2(W_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 = (W_2 W_1)\mathbf{x} + (W_2 \mathbf{b}_1 + \mathbf{b}_2)$ is still affine, however many layers there are, and an affine score yields linear decision boundaries. Hidden layers

are learned embeddings: e.g. $\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$ separates an inner from an outer circle by a plane. A network learns such combinations instead of receiving them by hand; depth composes them hierarchically (strokes $\to$ loops $\to$ digits). Depth increases expressiveness but does not by itself guarantee trainability or generalization (Chapters 6–8).

4 Activation Functions and Outputs

Activation	Definition	Derivative
Sigmoid	$\sigma(z) = (1 + e^{-z})^{-1} \in (0, 1)$	$\sigma(1 - \sigma) \in (0, 1/4]$
Tanh	$\tanh z = 2\sigma(2z) - 1 \in (-1, 1)$	$1 - \tanh^2 z \in (0, 1]$
ReLU	$\max(0, z)$	$\mathbf{1}\{z > 0\}$
Leaky ReLU	$\max(az, z)$, $0 < a < 1$	1 or a
ELU	z $(z \geq 0)$, $a(e^z - 1)$ $(z < 0)$	1 or ae^z
Maxout	$\max_k(\mathbf{w}_k^\top \mathbf{x} + b_k)$	routes to the max

Outputs: softmax for mutually exclusive classes (loss $-\log p_y$, score gradient $\mathbf{p} - \mathbf{e}_y$), sigmoid for binary probabilities, identity for regression.

Example 5.3: forward pass

$\mathbf{x} = (1, -2)$, $W_1 = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$, $W_2 = \mathbf{I}$, no biases, ReLU hidden layer: $\mathbf{z}_1 = (-1, 4)$, $\mathbf{a}_1 = (0, 4)$, scores $(0, 4)$, softmax $\approx (0.018, 0.982)$.

5 Computation Graphs and the Scalar Chain Rule

Training repeats: zero the gradients, forward pass, loss, backward pass (backpropagation computes derivatives), optimizer update. Backpropagation applies the chain rule on a computation graph: each node multiplies the *upstream* derivative (from later operations) by its *local* derivative.

Example 5.4: $f = (x + y)z$

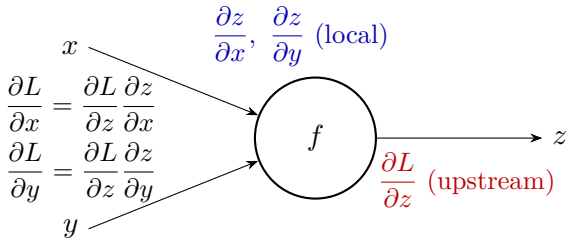
At $(x, y, z) = (-2, 5, -4)$: $q = x + y = 3$, $f = qz = -12$. Local derivatives $\partial f/\partial q = z = -4$, $\partial f/\partial z = q = 3$, $\partial q/\partial x = \partial q/\partial y = 1$, hence $\partial f/\partial x = \partial f/\partial y = -4$, $\partial f/\partial z = 3$.

Example 5.5: sigmoid neuron

$f = \sigma(w_0x_0 + w_1x_1 + w_2)$ with $w_0 = 2, x_0 = -1, w_1 = -3, x_1 = -2, w_2 = -3$: $z = 1$, $f \approx 0.731$, $\partial f/\partial z = f(1 - f) \approx 0.197$. Then $\partial f/\partial w_0 = 0.197x_0 = -0.197$, $\partial f/\partial x_0 = 0.197w_0 = 0.393$, $\partial f/\partial w_1 = -0.393$, $\partial f/\partial x_1 = -0.590$, $\partial f/\partial w_2 = 0.197$. Decomposing σ into $z \to -z \to e^{-z} \to 1 + e^{-z} \to 1/(\cdot)$ gives the same numbers; treating σ as one node is cheaper.

Remark 5.6: gate patterns

Add gate: distributes the upstream derivative to both inputs. Multiply gate: each input receives upstream $\times$ the other input. Copy (a value used twice): derivatives add. Max gate: routes the derivative to the maximal input. Hence a shared parameter accumulates gradients from all its uses.



Backpropagation at one node: the downstream gradient is the upstream gradient times the local gradient.

6 Vector Calculus Rules for Backpropagation

Motivation. Backpropagation is the chain rule organized so that each layer passes one vector backwards. A handful of vector rules (for linear maps, elementwise nonlinearities and losses) suffice to differentiate any feedforward network.

Gradient convention: $\nabla_{\mathbf{x}} L$ has the shape of $\mathbf{x}$; the Jacobian of $\mathbf{y} \in \mathbb{R}^m$ in $\mathbf{x} \in \mathbb{R}^n$ is $m \times n$ with entries $\partial y_i / \partial x_j$.

Theorem 5.7: rules R1–R6

R1. Sum rule: the Jacobian of a sum is the sum of Jacobians; terms not depending on the variable contribute 0.

R2. Identity: $\partial \mathbf{x} / \partial \mathbf{x} = \mathbf{I}$.

R3. Chain rule: $\frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \frac{\partial \mathbf{z}}{\partial \mathbf{y}} \frac{\partial \mathbf{y}}{\partial \mathbf{x}}$ (outer Jacobian on the left).

R4. Affine map: $\partial(A\mathbf{x} + \mathbf{b}) / \partial \mathbf{x} = A$.

R5. Elementwise map: $\partial \sigma(\mathbf{z}) / \partial \mathbf{z} = \operatorname{diag}(\sigma'(\mathbf{z}))$, so $\operatorname{diag}(\sigma'(\mathbf{z}))\mathbf{v} = \sigma'(\mathbf{z}) \odot \mathbf{v}$.

R6. Gradient chain rule (scalar L): $\nabla_{\mathbf{x}} L = \left(\frac{\partial \mathbf{u}}{\partial \mathbf{x}}\right)^{\top} \nabla_{\mathbf{u}} L$.

Proof. R6: by R3 the row vector $\partial L / \partial \mathbf{x} = (\partial L / \partial \mathbf{u})(\partial \mathbf{u} / \partial \mathbf{x})$; transposing reverses the order. ■

Theorem 5.8: affine layer

For $\mathbf{z} = W\mathbf{a} + \mathbf{b}$ and $\boldsymbol{\delta} = \nabla_{\mathbf{z}} L$:

$$\nabla_W L = \boldsymbol{\delta}\, \mathbf{a}^{\top}, \qquad \nabla_{\mathbf{b}} L = \boldsymbol{\delta}, \qquad \nabla_{\mathbf{a}} L = W^{\top} \boldsymbol{\delta}.$$

For an activation layer $\mathbf{h} = \sigma(\mathbf{z})$: $\nabla_{\mathbf{z}} L = \nabla_{\mathbf{h}} L \odot \sigma'(\mathbf{z})$.

Proof. $\nabla_{\mathbf{a}} L$: R6 with R4. $\nabla_{\mathbf{b}} L$: R6 with $\partial \mathbf{z} / \partial \mathbf{b} = \mathbf{I}$ (R1, R2). $\nabla_W L$: row i of W, $\mathbf{w}_i^{\top}$, affects L only through $z_i = \mathbf{w}_i^{\top} \mathbf{a} + b_i$, whose gradient in $\mathbf{w}_i$ is $\mathbf{a}$ (R4); thus $\nabla_{\mathbf{w}_i} L = \delta_i \mathbf{a}$, and stacking rows gives $\boldsymbol{\delta} \mathbf{a}^{\top}$. Equivalently, componentwise $\partial L / \partial W_{ij} = \delta_i a_j$. Activation: R6 with R5, and a diagonal matrix is symmetric. ■

Remark 5.9: differentials and trace

Matrix gradients can also be read from differentials: if $dL = \operatorname{tr}(K\, dW)$ then $\nabla_W L = K^{\top}$. For $\mathbf{z} = W\mathbf{a}$, $dL = \boldsymbol{\delta}^{\top}(dW)\mathbf{a} = \operatorname{tr}(\mathbf{a}\boldsymbol{\delta}^{\top} dW)$, so $\nabla_W L = \boldsymbol{\delta}\mathbf{a}^{\top}$.

6.1 Backpropagation through an MLP

With $\boldsymbol{\delta}_\ell = \nabla_{\mathbf{z}^{(\ell)}} L$ (an $n_\ell \times 1$ column), the recursion is

$$\boldsymbol{\delta}_\ell = (W_{\ell+1}^\top \boldsymbol{\delta}_{\ell+1}) \odot \sigma'(\mathbf{z}^{(\ell)}), \qquad \nabla_{W_\ell} L = \boldsymbol{\delta}_\ell \, \mathbf{a}^{(\ell-1)\top}, \qquad \nabla_{\mathbf{b}_\ell} L = \boldsymbol{\delta}_\ell.$$

It follows from applying R6 twice: through the affine map ($W_{\ell+1}^\top$) and through the activation (elementwise product). In a batch, outer products are summed (or averaged) over samples.

Example 5.10: forward example continued backward

Take class 1 as correct: $\boldsymbol{\delta}_2 = \mathbf{p} - \mathbf{e}_1 \approx (-0.982, 0.982)$, $\nabla_{W_2} L = \boldsymbol{\delta}_2 \mathbf{a}_1^\top = \begin{pmatrix} 0 & -3.928 \\ 0 & 3.928 \end{pmatrix}$. The first hidden pre-activation was negative, so $\boldsymbol{\delta}_1 = (0, 0.982)$ and $\nabla_{W_1} L = \boldsymbol{\delta}_1 \mathbf{x}^\top = \begin{pmatrix} 0 & 0 \\ 0.982 & -1.964 \end{pmatrix}$: the inactive unit receives no gradient (a "dead ReLU" for this input). For elementwise ReLU, upstream $(4, -1, 5, 9)$ at $\mathbf{x} = (1, -2, 3, -1)$ becomes $(4, 0, 5, 0)$ with no Jacobian ever formed.

7 Worked Results

Motivation. Two standard computations show what the rules reveal: why cross-entropy trains a sigmoid neuron faster than squared error, and how a residual connection keeps the backward signal from vanishing.

7.1 Single neurons: MSE versus cross-entropy

Theorem 5.11: gradient bounds

For $y = \sigma(wx + b)$ and $C = \frac{1}{2}(y - z)^2$: $\nabla C = y(1-y)(y-z)(x, 1)$ and $\|\nabla C\| < \frac{1}{4}\sqrt{1 + x^2}\,(1 + |z|)$. For cross-entropy $C = -[z \log y + (1 - z) \log(1 - y)]$: $\nabla C = (y - z)(x, 1)$.

Proof. Chain rule with $\sigma' = y(1 - y) \le 1/4$ and $|y - z| \le |y| + |z| < 1 + |z|$ since $0 < y < 1$. For cross-entropy, $\partial C / \partial t = -[z(1 - y) - (1 - z)y] = y - z$: the factor σ' cancels. ∎

For a saturated wrong neuron ($z = 1$, $wx + b = -10$), the MSE gradient carries the factor $\sigma' \approx 4.5 \times 10^{-5}$ while the cross-entropy gradient is of order one: MSE with sigmoid suffers a learning slowdown. GD reads $(w_{n+1}, b_{n+1}) = (w_n, b_n) - \eta \nabla C$.

7.2 Residual MLP

Consider $y_\ell = \sigma(A_\ell y_{\ell-1} + b_\ell) + y_{\ell-1}$ for $2 \le \ell \le L - 1$, $y_1 = \sigma(A_1 x + b_1)$, $y_L = A_L y_{L-1} + b_L$ (scalar). Put $z_\ell = A_\ell y_{\ell-1} + b_\ell$, $D_\ell = \operatorname{diag}(\sigma'(z_\ell))$ and $g_\ell = \partial y_L / \partial y_\ell \in \mathbb{R}^{1 \times m}$.

Theorem 5.12: residual Jacobians and gradients

$$\frac{\partial y_\ell}{\partial y_{\ell-1}} = I + D_\ell A_\ell \ (2 \le \ell \le L - 1), \qquad \frac{\partial y_L}{\partial y_{L-1}} = A_L, \qquad g_\ell = A_L \prod_{j=L-1}^{\ell+1} (I + D_j A_j),$$

$$\frac{\partial y_L}{\partial b_\ell} = g_\ell D_\ell, \qquad \frac{\partial y_L}{\partial A_\ell} = D_\ell g_\ell^\top y_{\ell-1}^\top \quad \left(\tfrac{\partial y_L}{\partial (A_\ell)_{ij}} = (g_\ell)_i \sigma'((z_\ell)_i)(y_{\ell-1})_j\right),$$

and $\partial y_L/\partial b_L = 1$, $\partial y_L/\partial A_L = y_{L-1}^\top$.

Proof. R1 splits y_ℓ into $\sigma(z_\ell)$ and $y_{\ell-1}$; R2 gives I; R3 with R5, R4 gives $D_\ell A_\ell$. For b_ℓ, the residual term does not depend on b_ℓ and $\partial z_\ell/\partial b_\ell = I$, so $\partial y_\ell/\partial b_\ell = D_\ell$; chain with g_ℓ. For A_ℓ, the ith row affects y_L only through $(z_\ell)_i$, giving the componentwise formula; it factorizes into an i-part and a j-part, i.e. an outer product. ■

If $A_j = 0$ or $\sigma'(z_j) = 0$ for some intermediate j, the factor is $I + 0 = I$, not 0: the identity path carries the gradient, so lower-layer gradients need not vanish. Without the residual, the factor is $D_j A_j = 0$ and all lower gradients are exactly zero. Expanding $\prod(I + D_j A_j) = I + \sum D_j A_j + \cdots$ exhibits the skip path explicitly (Chapter 7).

8 Selected exercises

Motivation. The exercises prove that a hidden layer is genuinely needed for XOR, derive a conservation law that the backpropagation rules impose on two-layer linear networks, and justify the finite-difference test used to check a backward pass.

Exercise 5.1 (XOR needs a hidden layer)

Let $A = \{(0,0),(1,1)\}$ and $B = \{(0,1),(1,0)\}$.
(a) Show that no affine $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$ satisfies $f > 0$ on B and $f < 0$ on A, nor the reverse.
(b) Show that $g(\mathbf{x}) = \mathrm{ReLU}(x_1 + x_2) - 2\,\mathrm{ReLU}(x_1 + x_2 - 1)$ equals 0 on A and 1 on B.
(c) Show that $\{\mathbf{x} \in \mathbb{R}^2 : g(\mathbf{x}) > \frac{1}{2}\}$ is the open strip $\{\frac{1}{2} < x_1 + x_2 < \frac{3}{2}\}$, which is not a half-plane.

Solution. [Supplement]

(a) Suppose $f > 0$ on B and $f < 0$ on A. From B: $w_2 + b > 0$ and $w_1 + b > 0$, so $w_1 + w_2 + 2b > 0$. From A: $b < 0$ and $w_1 + w_2 + b < 0$, so $w_1 + w_2 + 2b < 0$, a contradiction. The reverse case is the same argument applied to $-f$.

(b) g depends on $\mathbf{x}$ only through $t = x_1 + x_2$, via $h(t) = \mathrm{ReLU}(t) - 2\,\mathrm{ReLU}(t-1)$. On A, $t \in \{0, 2\}$ and $h(0) = 0$, $h(2) = 2 - 2 = 0$; on B, $t = 1$ and $h(1) = 1$.

(c) Piecewise, $h(t) = 0$ for $t \le 0$, $h(t) = t$ for $0 \le t \le 1$, and $h(t) = 2 - t$ for $t \ge 1$. Hence $h(t) > \frac{1}{2}$ iff $\frac{1}{2} < t < \frac{3}{2}$. The complement of a half-plane is a half-plane, hence convex. The complement of the strip contains $(0,0)$ and $(1,1)$ but not their midpoint $(\frac{1}{2}, \frac{1}{2})$, so it is not convex and the strip is not a half-plane. Two ReLU units with fixed weights already produce a non-linear decision boundary. ■

Exercise 5.2 (A conserved quantity of two-layer linear networks)

Let $W_1 \in \mathbb{R}^{h \times n}$, $W_2 \in \mathbb{R}^{m \times h}$, and $L(W_1, W_2) = \ell(W_2 W_1)$ for a differentiable $\ell : \mathbb{R}^{m \times n} \to \mathbb{R}$ with gradient $G = \nabla \ell(M)$ at $M = W_2 W_1$ (so $d\ell = \mathrm{tr}(G^\top dM)$). Let $Q = W_1 W_1^\top - W_2^\top W_2 \in \mathbb{R}^{h \times h}$.
(a) Show that $\nabla_{W_1} L = W_2^\top G$ and $\nabla_{W_2} L = G W_1^\top$.
(b) Show that along the gradient flow $\dot{W}_k = -\nabla_{W_k} L$ $(k = 1, 2)$, Q is constant in time.
(c) Show that one gradient descent step with step η changes Q by exactly $\eta^2 (A A^\top - B^\top B)$, where $A = \nabla_{W_1} L$ and $B = \nabla_{W_2} L$.

Solution. [Supplement]

(a) By the product rule $dM = (dW_2)W_1 + W_2\,dW_1$, so by cyclicity of the trace

$$dL = \operatorname{tr}\left(G^\top dW_2\, W_1\right) + \operatorname{tr}\left(G^\top W_2\, dW_1\right) = \operatorname{tr}\left((W_1 G^\top)\, dW_2\right) + \operatorname{tr}\left((G^\top W_2)\, dW_1\right).$$

By the differential rule of this chapter ($dL = \operatorname{tr}(K\,dW) \Rightarrow \nabla_W L = K^\top$), $\nabla_{W_2} L = G W_1^\top$ and $\nabla_{W_1} L = W_2^\top G$.

(b) By the product rule and (a),

$$\frac{d}{dt}(W_1 W_1^\top) = \dot W_1 W_1^\top + W_1 \dot W_1^\top = -W_2^\top G W_1^\top - W_1 G^\top W_2,$$
$$\frac{d}{dt}(W_2^\top W_2) = \dot W_2^\top W_2 + W_2^\top \dot W_2 = -W_1 G^\top W_2 - W_2^\top G W_1^\top.$$

The two right-hand sides are equal, so $\dot Q = 0$.

(c) With $W_1' = W_1 - \eta A$ and $W_2' = W_2 - \eta B$,

$$W_1' W_1'^\top = W_1 W_1^\top - \eta(A W_1^\top + W_1 A^\top) + \eta^2 A A^\top,$$
$$W_2'^\top W_2' = W_2^\top W_2 - \eta(B^\top W_2 + W_2^\top B) + \eta^2 B^\top B.$$

By (a), $A W_1^\top = W_2^\top G W_1^\top = W_2^\top B$ and $W_1 A^\top = W_1 G^\top W_2 = B^\top W_2$, so the first-order terms cancel and $Q' - Q = \eta^2(AA^\top - B^\top B)$. If the layers start balanced ($Q = 0$, e.g. $W_1 = W_2^\top$ when $m = n$), gradient flow keeps them balanced, and small-step gradient descent keeps them nearly so. The rescaling $W_1 \mapsto cW_1$, $W_2 \mapsto c^{-1}W_2$ leaves $W_2 W_1$ unchanged but changes Q, so gradient flow never moves along it. ■

Exercise 5.3 (Checking a backward pass by finite differences)

Let $\varphi : \mathbb{R} \to \mathbb{R}$ be three times continuously differentiable on $[x-h, x+h]$, $h > 0$, and $M_k = \max_{[x-h,x+h]} |\varphi^{(k)}|$.
(a) Show that $\left|\frac{\varphi(x+h)-\varphi(x)}{h} - \varphi'(x)\right| \le \frac{h}{2} M_2$.
(b) Show that $\frac{\varphi(x+h)-\varphi(x-h)}{2h} - \varphi'(x) = \frac{h^2}{6}\varphi'''(\xi)$ for some $\xi \in [x-h, x+h]$, so the error is at most $\frac{h^2}{6} M_3$.
(c) For $\varphi = \exp$ at $x = 0$ and $h = 10^{-3}$, compute both errors to leading order. Explain how this checks $\partial L/\partial \theta_j$ computed by backpropagation.

Solution. [Supplement]

(a) Taylor's theorem with Lagrange remainder gives $\varphi(x+h) = \varphi(x) + h\varphi'(x) + \frac{h^2}{2}\varphi''(\xi)$ with $\xi \in (x, x+h)$. Rearranging, the error equals $\frac{h}{2}\varphi''(\xi)$, bounded by $\frac{h}{2} M_2$.

(b) *Step 1 (two expansions).* To third order,

$$\varphi(x \pm h) = \varphi(x) \pm h\varphi'(x) + \frac{h^2}{2}\varphi''(x) \pm \frac{h^3}{6}\varphi'''(\xi_\pm), \qquad \xi_+ \in (x, x+h),\ \xi_- \in (x-h, x).$$

Step 2 (subtract). The even terms cancel: $\varphi(x+h) - \varphi(x-h) = 2h\varphi'(x) + \frac{h^3}{6}(\varphi'''(\xi_+) + \varphi'''(\xi_-))$. Dividing by $2h$, the error is $\frac{h^2}{6} \cdot \frac{1}{2}(\varphi'''(\xi_+) + \varphi'''(\xi_-))$.

Step 3 (one point). The average of two values of the continuous φ''' lies between them, so by the intermediate value theorem it equals $\varphi'''(\xi)$ for some ξ between ξ_- and ξ_+.

(c) Here $\varphi'(0) = 1$. The forward quotient is $(e^h - 1)/h = 1 + \frac{h}{2} + \frac{h^2}{6} + \cdots$, with error about 5.0×10^{-4}; the central quotient is $\sinh(h)/h = 1 + \frac{h^2}{6} + \cdots$, with error about 1.7×10^{-7}. To check

backpropagation, apply (b) to $\varphi(t) = L(\boldsymbol{\theta} + t\mathbf{e}_j)$, whose derivative at $t = 0$ is $\partial L/\partial\theta_j$: two forward passes at $\boldsymbol{\theta} \pm h\mathbf{e}_j$ give an independent estimate accurate to $O(h^2)$, to be compared (in relative error) with the backpropagated value. In floating point h cannot be taken too small, since rounding errors of size about ϵ_{mach}/h then dominate. ■